ESE 2150 Lab 5

Voltage Follower, Cascaded Amplifiers, and a 3-Bit DAC

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A. Op-Amp as a Voltage Follower

1. Procedure and Observations

We built a voltage follower on a breadboard using one op-amp in the LM324AN chip. To provide $+5\,\mathrm{V}$ and $-5\,\mathrm{V}$, we set two power-supply channels to $5\,\mathrm{V}$. We connected the negative terminal of the first channel to the positive terminal of the second and treated this shared connection as ground. The remaining positive terminal provided $+5\,\mathrm{V}$, and the remaining negative terminal provided $-5\,\mathrm{V}$. We connected these supplies to pins 4 and 11, respectively.

We connected the op-amp output directly to its inverting input and connected the trimmer's adjustable voltage to the non-inverting input. This negative-feedback connection makes the output approximately match the input. We adjusted the trimmer to the listed input voltages and measured the input and output relative to the same ground.

From 0 to $3.5\,\mathrm{V}$, the output closely followed the input. For example, a $1\,\mathrm{V}$ input produced a $0.98\,\mathrm{V}$ output, while inputs of $2.5\,\mathrm{V}$ and $3.5\,\mathrm{V}$ produced matching outputs. At higher inputs, the output stopped increasing and remained near $3.75\,\mathrm{V}$.

This plateau indicates output saturation. The LM324AN requires headroom below its positive supply rail, so powering it with $5\,\mathrm{V}$ does not mean it can produce a $5\,\mathrm{V}$ output under these conditions.

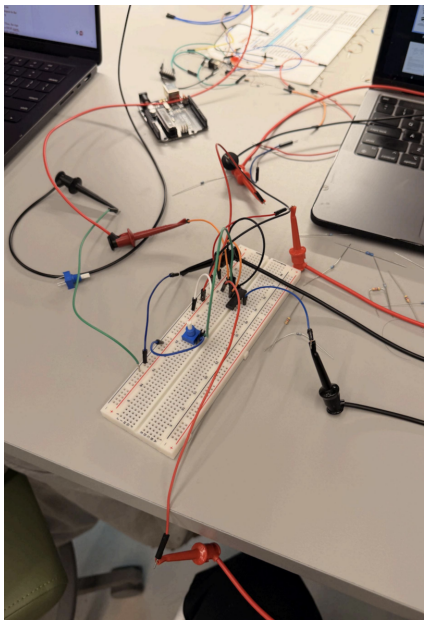


Figure 1: Voltage-follower setup.

2. Results and Error

The handout defines signed percentage error as

$$E = 100\frac{V_o - V_p}{V_p}.$$

At $V_p = 0$, this expression is undefined.

V_p (V)	V_o (V)	Signed error (%)
0.0	0.000	N/A
0.5	0.520	4.00
1.0	0.980	-2.00
1.5	1.480	-1.33
2.0	2.053	2.65
2.5	2.500	0.00
3.0	3.000	0.00
3.5	3.500	0.00
4.0	3.750	-6.25
4.5	3.750	-16.67
4.8	3.750	-21.88

Table 1: Voltage-follower measurements.

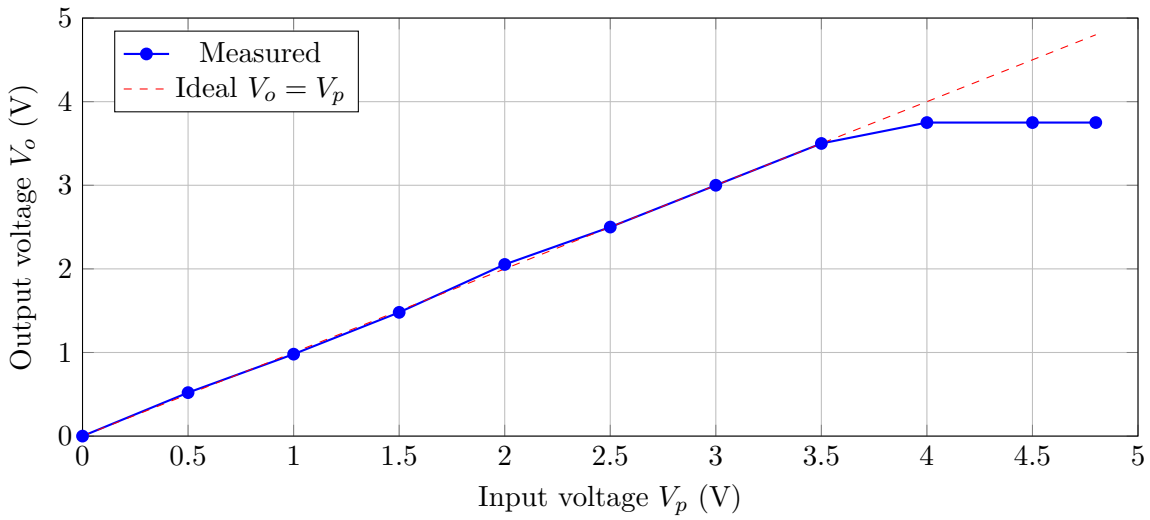


Figure 2: The buffer follows the input at lower voltages and saturates near 3.75 V.

The output approximately follows the input from 0 to 3.5 V. In this range, the error magnitude is at most 4%, consistent with measurement uncertainty and nonideal op-amp behavior.

At inputs of 4 V and above, the output is limited to approximately 3.75 V. The error therefore increases with input voltage, reaching a magnitude of 21.88% at 4.8 V.

B. Non-Inverting–Inverting Amplifier

1. Procedure and Observations

We built the two-stage amplifier circuit using two op-amps in the LM324AN package and powered it with +5 V and −5 V. The first stage was non-inverting, and the second stage was inverting.

We applied a 1 kHz sinusoidal input and monitored the input, first-stage output, and final output using oscilloscope Channels 1, 2, and 3. We first used a nominal input amplitude of 0.5 V, then repeated the measurement at 2 V.

At the lower amplitude, the first-stage output was in phase with the input and approximately twice its size. The final output was inverted and approximately 4.38 times the input magnitude. The signals remained approximately sinusoidal.

At the higher amplitude, the first-stage output showed positive clipping, and the final output was strongly clipped at both peaks.

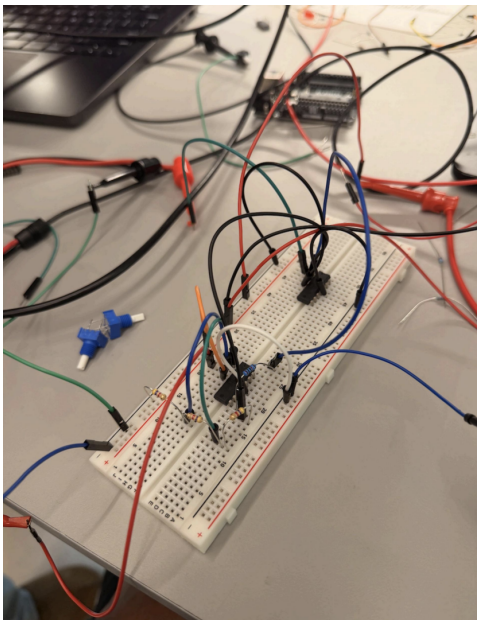


Figure 3: Cascaded amplifier setup.

2. Theoretical Gains

The first-stage gain is

$$A_1 = 1 + \frac{R_f}{R_g} = 1 + \frac{4.7}{4.7} = 2.$$

The second-stage gain is

$$A_2 = -\frac{10}{4.7} \approx -2.128.$$

The overall gain in the linear operating region is

$$A_{\text{system}} = A_1 A_2 = 2\left(-\frac{10}{4.7}\right) \approx -4.255.$$

3. Nominal Input Amplitude of $0.5\,\mathrm{V}$



Figure 4: Measured amplifier waveforms at a nominal $0.5\,\mathrm{V}$ input amplitude and $1\,\mathrm{kHz}$.

a. Non-Inverting Stage

The input and first-stage output had the same frequency and were in phase. Their measured peak-to-peak voltages were

$$V_{\mathrm{in,pp}} = 1.07\,\mathrm{V}, \qquad V_{\mathrm{stage1,pp}} = 2.14\,\mathrm{V}.$$

Therefore,

$$A_{1,\mathrm{measured}} = \frac{2.14}{1.07} = 2.00.$$

This agrees with the theoretical gain of 2 to the displayed precision. The output remained sinusoidal without visible clipping, consistent with approximately linear operation at this amplitude.

The channels used different vertical scales, so their displayed heights should not be compared without accounting for those scales.

b. Overall System

The final output had the same frequency as the input but was approximately 180° out of phase. Using the measured peak-to-peak voltages and the observed inversion,

$$A_{\mathrm{system,measured}} = -\frac{4.69}{1.07} \approx -4.383.$$

The theoretical gain is approximately -4.255, so the measured gain magnitude was higher by

$$\frac{4.383 - 4.255}{4.255} \times 100\% \approx 3.0\%.$$

This difference could result from resistor tolerances, measurement uncertainty, and nonideal op-amp behavior. The final output remained approximately sinusoidal without visible clipping, so the system operated approximately linearly at this input amplitude.

4. Nominal Input Amplitude of 2 V

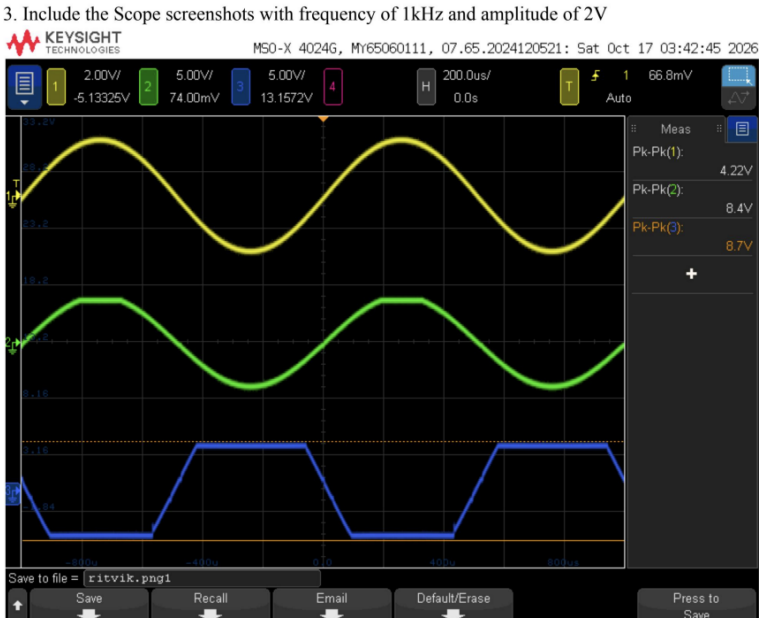


Figure 5: Measured amplifier waveforms at a nominal 2 V input amplitude and 1 kHz.

a. Non-Inverting Stage

The first-stage output remained in phase with the input and had the same repetition frequency. The measured peak-to-peak ratio was

$$\frac{V_{\text{stage1,pp}}}{V_{\text{in,pp}}} = \frac{8.4}{4.22} \approx 1.99.$$

Although this ratio is close to the theoretical gain of 2, the positive peaks were visibly flattened. The output had reached its upper voltage limit, so the amplifier was not linear over the full waveform. A nearly correct peak-to-peak ratio does not account for the waveform distortion.

b. Overall System

The final output was inverted and heavily clipped at both peaks. Its signed peak-to-peak amplitude ratio was

$$-\frac{V_{\text{out,pp}}}{V_{\text{in,pp}}} = -\frac{8.7}{4.22} \approx -2.06.$$

This magnitude is much smaller than the theoretical linear gain of approximately 4.26 because the required output exceeds the available output voltage swing.

The system was not linear at this amplitude. Its output was clipped rather than remaining a scaled, inverted copy of the input. Therefore, -2.06 describes the measured amplitude ratio and should not be treated as a constant linear gain.

C(a). Manually Controlled 3-Bit R-2R DAC

1. Procedure and Observations

We built the three-bit R-2R DAC using $3.3\,\mathrm{k\Omega}$ and $6.8\,\mathrm{k\Omega}$ resistors and one LM324AN op-amp, powered by $+5\,\mathrm{V}$ and $-5\,\mathrm{V}$.

We manually changed the three inputs through the binary combinations from 000 to 111. Each bit was connected to ground for 0 or $+5\,\mathrm{V}$ for 1. We measured the output relative to ground for each combination.

The output became more negative as the binary input increased, going from $0\,\mathrm{V}$ for 000 to $-4.311\,\mathrm{V}$ for 111. Most measured outputs were more negative than the reported calculated values.

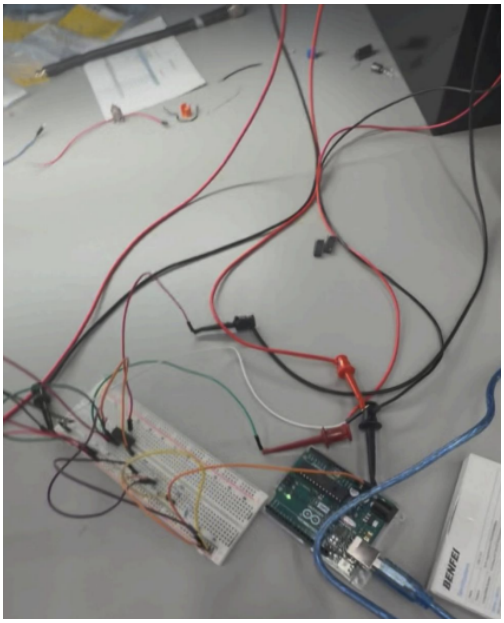


Figure 6: Manually controlled DAC setup.

2. Calculations

The original hand calculations used superposition, activating one digital input at a time while grounding the others. The reported individual output contributions were

$$V_{\mathrm{LSB}} \approx -0.623\,\mathrm{V},$$
$$V_{\mathrm{middle}} \approx -1.227\,\mathrm{V},$$
$$V_{\mathrm{MSB}} \approx -2.426\,\mathrm{V}.$$

For the MSB alone, the submitted calculation was

$$I_{\mathrm{MSB}} = \frac{5\,\mathrm{V}}{6.8\,\mathrm{k\Omega}} \approx 0.7353\,\mathrm{mA},$$

giving

$$V_{\mathrm{MSB}} = -(3.3\,\mathrm{k\Omega})I_{\mathrm{MSB}} \approx -2.426\,\mathrm{V}.$$

Using the reported bit contributions,

$$V_{\mathrm{out}} \approx -2.426b_2 - 1.227b_1 - 0.623b_0,$$

where b_2, b_1, and b_0 denote the MSB, middle bit, and LSB, respectively.

The output calculations are

$$\begin{aligned}
V(000) &= 0,\\
V(001) &\approx -0.623\,\mathrm{V},\\
V(010) &\approx -1.227\,\mathrm{V},\\
V(011) &\approx -1.849\,\mathrm{V},\\
V(100) &\approx -2.426\,\mathrm{V},\\
V(101) &\approx -3.049\,\mathrm{V},\\
V(110) &\approx -3.653\,\mathrm{V},\\
V(111) &\approx -4.276\,\mathrm{V}.
\end{aligned}$$

3. Complete Results Table

Bits	Decimal	Measured V_o (V)	Reported calculated V_o (V)
000	0	0.000	0.000
001	1	-0.738	-0.623
010	2	-1.449	-1.227
011	3	-2.118	-1.849
100	4	-2.870	-2.426
101	5	-3.609	-3.049
110	6	-3.902	-3.653
111	7	-4.311	-4.276

Table 2: Measured and originally reported calculated DAC outputs.

4. Output Plot and Discussion

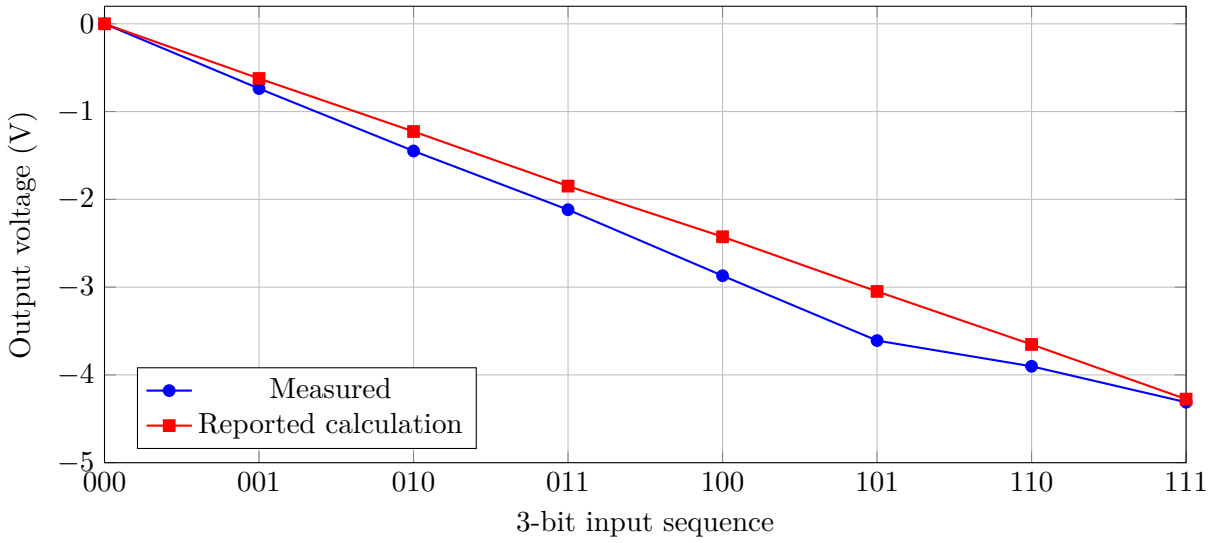


Figure 7: DAC output versus binary input code.

The output generally became more negative as the binary input increased, consistent with an inverting DAC. Larger input codes produce more current through the feedback resistor and therefore a more negative output.

Most measured outputs were more negative than the calculated values. Possible contributors include differences between actual and assumed resistor values, a high input voltage different from exactly 5 V, measurement uncertainty, and nonideal op-amp behavior.

The voltage steps became smaller near 110 and 111. This could be consistent with limited output swing, but the measurements alone do not establish the cause. Wiring and calculation accuracy would also need to be checked.

C(b). Arduino-Controlled 3-Bit DAC

1. Procedure

We connected the DAC inputs to Arduino UNO digital pins 8, 7, and 6 for the MSB, middle bit, and LSB, respectively. We powered the LM324AN with +5 V and −5 V supplies and connected the Arduino ground to the circuit ground.

We uploaded code intended to repeatedly generate 000, 001, 010, 011, 100, 101, 110, and 111. We monitored the DAC output using oscilloscope Channel 1, first with a 1 ms delay between codes and then with a 2 ms delay.

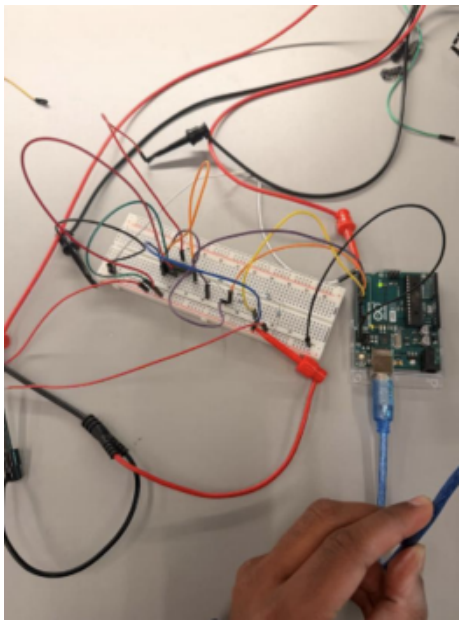


Figure 8: Arduino-controlled DAC setup.

2. Output with a 1 ms Delay

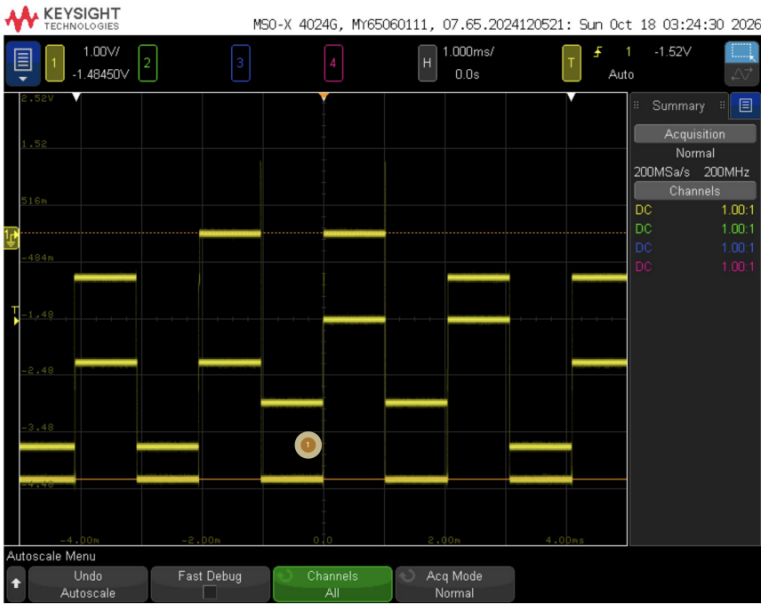


Figure 9: Submitted oscilloscope capture with `pause=1`. The captured waveform does not match the required descending staircase.

The expected waveform is an eight-level descending staircase. Each successive code should produce a more negative voltage, followed by a return to the 000 output when the sequence repeats.

Ignoring code-execution overhead, the expected sequence period is

$$T_1 \approx 8(1\,\text{ms}) = 8\,\text{ms}.$$

However, the submitted screenshot does not show eight successive descending levels. Its output jumps between levels in a different order. Therefore, the screenshot does not verify correct DAC operation.

The code sequence, Arduino pin assignments, ladder connections, and shared ground should be checked before repeating the measurement. The screenshot alone does not identify the specific fault.

3. Output with a 2 ms Delay

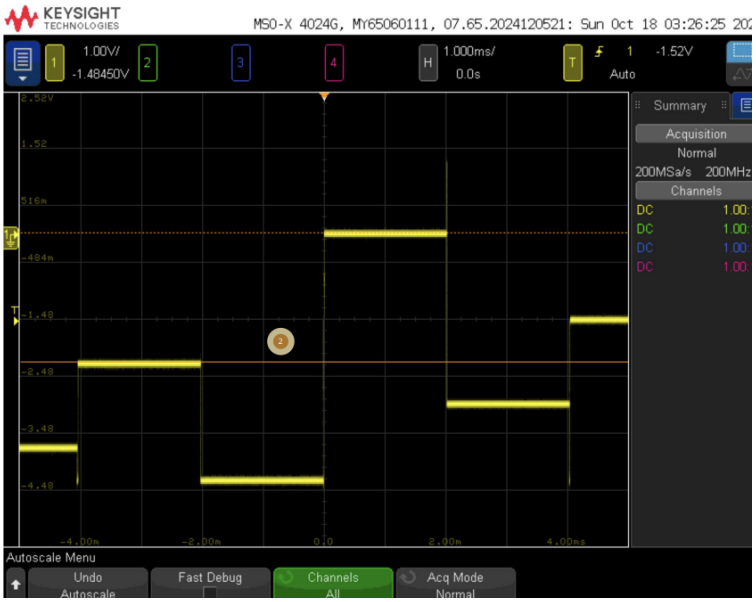


Figure 10: Submitted oscilloscope capture with `pause=2`. The longer plateaus are visible, but the output sequence remains incorrect.

Increasing the programmed delay from 1 ms to 2 ms should approximately double each plateau's duration while leaving its voltage level unchanged. The expected sequence period becomes

$$T_2 \approx 8(2\,\text{ms}) = 16\,\text{ms}.$$

The second screenshot shows longer horizontal plateaus, consistent with a longer delay. However, it still does not show the required descending staircase. A change in timing does not establish that the digital sequence is mapped correctly to the DAC inputs.